

ARC Lecture Bank — Quiz Results

Score: 30 / 76 (39.5%)

Topics: PD_EqpDes

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#	Question	Your Answer	Correct Answer	Result	Time (s)
1	Le/D for couplings and unions would be	200	negligible	Wrong	5.3
2	Normally; the length to diameter ratio of rotary dryers varies from 4 to 10 and for its economic operation, the number of transfer units (NTU) for such dryers varies from	3.5-5.5	1.5-2.5	Wrong	60.2
3	When one of the fluids is highly corrosive and has fouling tendency; it should	preferably flow inside the tube for its easier internal cleaning;	preferably flow inside the tube for its easier internal cleaning;	Correct	37.4
4	The ideal size of round bubble caps to be used in industrial distillation column having a diameter of 3-6 meters is _____ cm	15	15	Correct	19.3
5	The minimum plate spacing in most of the petroleum refinery distillation columns (of $D > 3\text{ft}$) is normally _____ inches	18	18	Correct	13.9
6	A certain pressure vessel manufacturer avoids reinforcements calculations for openings by always providing a reinforcing extending up to double the diameter of the opening and of the same material and thickness as that of the shell wall. If area compensation is accepted as a code guideline, his approach leads to safe design	irrespective of the shape of the vessel	irrespective of the shape of the vessel	Correct	35.5
7	A 'rupture disc' is provided in chemical equipment as an accessory meant for	relieving excess pressure	relieving excess pressure	Correct	10.9
8	Back-trapping in a distillation column results due to	excessive liquid gradient over the tray;	excessive liquid gradient over the tray;	Correct	24.4
9	Equivalent diameter for heat transfer calculation for a duct of square cross section having each side as 'd' will be	$\sqrt[4]{2}d$	d	Wrong	9.3
10	In a vertical short tube evaporator (Calendria type), the	area of central downtake is equal to 40 to 100% of total cross-sectional area of the surrounding tube	all (a); (b) & (c):	Wrong	20.0
11	In inclined long tube vertical evaporators, the tubes are inclined at an angle of	60 deg	45 deg	Wrong	12.7
12	The longitudinal stress induced in a thin-walled cylindrical vessel under internal pressure is	$pd/4t$	$pd/4t$	Correct	27.5
13	The normal range of velocity of water in pipes is _____ m/s	0.1-0.5	1-2	Wrong	6.7
14	In case of plain carbon steel, butt welded joints are used for shell plate thickness < _____ cm	0.5	0.5	Correct	19.4
15	A pipe is generally made of circular cross-section; because a circular cross-section has the	both (a) & (b)	both (a) & (b)	Correct	18.7
16	In the case of a shell and tube heat exchanger; the logarithmic mean temperature difference	is always less than arithmetic average value	is always less than arithmetic average value	Correct	42.5
17	For a given design of bubble cap, the number of bubble caps to be used per tray is set by the	diameter of the column	allowable gas velocity through the slots.	Wrong	327.0
18	Maximum size of the Raschig rings used in a packed tower is about _____ of the column diameter	1/10th	1/20th	Wrong	3.2
19	Normally; the ratio of the total riser area to the tower cross sectional area (for bubble cap towers of $D > 3\text{ft}$) is around	0.35-0.75	0.1-0.2	Wrong	7.5
20	The thermal stress in a metallic bar does not depend upon	both (a) and (b)	cross-sectional area	Wrong	6.7
21	Heat transfer co-efficient for a horizontal condenser as compared to a vertical condenser operating under similar	less	about 3 times	Wrong	9.6

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	conditions is				
22	In the allocated cap area, bubble caps are generally arranged on equilateral triangular pitch: Number of caps fixed on a plate is with a	cap pitch of 1.3-2 times the cap diameter	either 'a' or 'b'	Wrong	230.5
23	In case of saddle packing (e.g., berl and intralox saddles), the maximum size of the packing should not exceed of the column diameter	1/15th	1/15th	Correct	27.5
24	Any shell opening greater than 5 cms for a storage tank must be reinforced for reason of	preventing the local overstressing of the shell around the opening	preventing the local overstressing of the shell around the opening	Correct	7.2
25	Stage efficiency for packed tower varies with the	operating pressure and column diameter	all (a), (b) and (c):	Wrong	14.0
26	Large process vessels operating under extreme temperature and pressure conditions are normally clad/ lined with materials like stainless steel, aluminum_ nickel & its 'alloys, copper & its alloys and titanium. The cladding thickness is generally about _____percent of the total thickness	10 to 20	10 to 20	Correct	8.5
27	Value of Peclet number = 0, is the representative of	laminar flow:	complete back mixing:	Wrong	20.7
28	Column support for the roof of cylindrical storage tank must be provided for	small diameter tall tanks	large diameter tanks	Wrong	20.8
29	The ratio of the largest load in a test to the original cross-sectional area of the test specimen is called the _____stress	yield point	ultimate	Wrong	12.4
30	The safe distance of habitation from a hazardous chemical plant (TLV of its product 1000 ppm) should be about _____kms	5	5	Correct	41.4
31	Smoker's equation for the calculation of number of equilibrium stages in a continuous binary distillation column is used, when the	relative volatility is close to one (e.g., separation of close boiling isomers)	relative volatility is close to one (e.g., separation of close boiling isomers)	Correct	9.8
32	In a double pipe (concentric) heat exchanger; the hydraulic radius for heat transfer (for a fluid flowing through the annulus) would be	D2 - D1 (D1 and D2 are I.D. of inner and outer pipes)	more than that for fluid flow:	Wrong	11.7
33	The function of manholes provided in the shell of a distillation column is to	give access to the individual trays for cleaning, maintenance and installation.	give access to the individual trays for cleaning, maintenance and installation.	Correct	10.7
34	Minimum tube pitch recommended for shell and tube heat exchangers is about _____times the outside diameter of the tube:	1.25	1.25	Correct	10.0
35	_____tower is the most suitable gas-liquid contacting device involving highly corrosive fluids	Sieve plate	Packed	Wrong	7.9
36	In case of a shell and tube heat exchanger; the effect of L/D on inside film heat transfer co-efficient (hi) vanishes after a Reynold number of	3000	10000	Wrong	18.7
37	Lapjoints are preferred for plates having thickness _____mm	'>5	'> 3	Wrong	4.5
38	A cylindrical storage tank can have a self-supported conical roof	if its diameter is more than 50 meters.	if its diameter is less than 15-20 meters.	Wrong	13.8
39	Which of the following lengths of heat exchanger tubes are normally not used practice?	6 meters	0.5 meter	Wrong	39.4
40	In a distillation column, minimum clearance to be maintained between the overflow weir and the last row of the bubble caps is _____cm	7.5	7.5	Correct	14.8
41	In a multipass shell and tube heat exchanger; the problem of differential expansion between the shell and tube passes is taken care of by using a	floating head tube sheet	either (a) or (b)	Wrong	2.4

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42	Typical static submergence for bubble cap plate column operating at atmospheric pressure may be around	0.5"	0.5"	Correct	14.6
43	Seamless pipe is made	from rolled strips formed into cylinders and seam-welded	both (b) and (c) rolling	Wrong	4.7
44	In packed absorption tower; onset of flooding usually occurs at a pressure drop of about _____mm of water column per meter height of packing:	200-250	50-75	Wrong	12.7
45	The length of straight rectangular weir used on cross-flow trays is generally _____the column diameter:	0.6-0.8 times	0.6-0.8 times	Correct	9.8
46	Friction factor for turbulent flow in a new pipe is given by	$f = 0.22 \text{ Nre}^{-0.5}$	$f = 0.04/\text{Nre}^{0.16}$	Wrong	13.0
47	Baffles may be eliminated for	low viscosity liquids (< 200 poise)	high viscosity liquids (> 600 poise)	Wrong	10.3
48	Skirt support is the most suitable for supporting _____vessels	large horizontal	tall vertical	Wrong	10.1
49	Liquid redistribution should be done in a packed tower packed with Raschig rings every 6 meters or _____ times the column diameter; whichever is lower:	5-7.5	2.5-3.0	Wrong	16.0
50	Pick out the wrong statement pertaining to a shell and tube heat exchanger	The maximum value of baffle spacing is equal to the I.D. of the shell	Disk and doughnut baffle is the most commonly used baffle	Wrong	14.6
51	Hoop (circumferential) stress induced in a thin walled 'Horton Sphere' used for the storage of liquid ammonia under pressure is	$pD/6t$	$pD/4t$	Wrong	4.1
52	Baffle spacing is generally _____ the ID of the shell	not less than one fifth of	both (b) and (c)	Wrong	18.2
53	Poisson's ratio of a material is the ratio of unit lateral strain to the unit axial elongation within its elastic limit: The value of Poisson's ratio for structural and pressure vessel steel may be taken as	0.3	0.3	Correct	48.9
54	In a shell and tube heat exchanger; the shortest center to center distance between the adjacent tubes is	called tube pitch:	called tube pitch:	Correct	13.2
55	The wall thickness of thin cylindrical shell with hemispherical ends is _____ that of the spherical ends	more than	more than	Correct	15.4
56	The minimum tray spacing in distillation column of diameter less than 3 ft is normally _____ inches	18	6	Wrong	11.8
57	The clearance between two tubes in a shell and tube heat exchanger is known as 'ligament'; whose minimum value is _____ the outside diameter of the tube; but in no case it should be less than 4.5 mm	half	one fourth	Wrong	20.7
58	Maximum plate efficiency is achieved in case of _____ flow:	cross	cross	Correct	5.4
59	What is the slope of the operating line in the rectifying section of a distillation column?	'>1'	<1	Wrong	13.5
60	The function of manholes provided in the shell of a distillation column is to	give access to the individual trays for cleaning, maintenance and installation.	give access to the individual trays for cleaning, maintenance and installation.	Correct	19.3
61	Generally; as the length of the liquid path on a tray is increased beyond 5 ft; the overall column efficiency	increases	increases	Correct	22.0
62	Removable connection of tubes to tube sheet is called	ferrule	ferrule	Correct	9.7
63	Diameter of bubble caps used in high pressure columns is _____ as compared to that used in identical vacuum columns:	less	less	Correct	55.4
64	The minimum baffle height should be	equal to the impeller diameter	twice the impeller diameter	Wrong	15.6

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65	For high pressure process equipment vessels, the connected nozzle should be	welded	welded	Correct	43.2
66	Slope of countercurrent flow rotary drier is in the range of _____ meters	0 to 1.5	0 to 0.08	Wrong	33.3
67	Pick out the wrong statement pertaining to the design of the distillation column.	If sufficient slot area is not provided, the gas may pass through the skirt clearance	None of these	Wrong	20.3
68	Joint efficiency (J) for a seamless pipe is	1.2	1	Wrong	2.7
69	At equal mass velocities, the values of film coefficients for gases as compared to that for liquids are generally:	same	lower	Wrong	24.6
70	The shell side pressure drop in a shell and tube heat exchanger is maximum for _____ baffle	segmental	orifice	Wrong	19.4
71	In distillation columns, bubble caps are held in place over trays by	rivets	both (b) and (c)	Wrong	9.4
72	In a continuous distillation column, the optimum reflux ratio ranges from times the minimum reflux ratio:	1.6 to 2	11 to 1.5	Wrong	10.0
73	Pick out the wrong statement about the design of sieve plate column and the bubble cap column:	Weir height in case of a bubble cap plate ranges from 50 to 150 mm and is higher than the sieve plate	Weir height for a bubble cap plate column is the same as that for a sieve plate column	Wrong	0.6
74	For a cylindrical internally pressurized vessel, which of the following closure types would withstand highest pressure, if each closure is of the same material and thickness?	hemispherical	hemispherical	Correct	11.6
75	Baffles are provided in a shell and tube heat exchanger to increase the turbulence and velocity of the shell side fluid. Which of the following shaped baffles does not fall in the category of transverse baffle?	Flat plate extending across the wall	Helical type baffle	Wrong	7.3
76	The ratio propeller agitator D/tank D is normally taken as	0.75-0.85	0.15-0.30	Wrong	4.4